- *Personas/profiles*—The primary persona and profile the agent will use. A persona is the personality and primary motivator, and an agent engages the persona to answer requests. We'll look in this chapter at how personas/profiles can be developed and consumed.
- *Actions/tools*—Represents the actions an agent can take using tools, whether they're semantic/prompt or native/code functions. In this chapter, we'll look at how to build both semantic and native functions within Nexus.
- *Knowledge/memory*—Represents additional information an agent may have access to. At the same time, agent memory can represent various aspects, from short-term to semantic memory.
- *Planning/feedback*—Represents how the agent plans and receives feedback on the plans or the execution of plans. Nexus will allow the user to select options for the type of planning and feedback an agent uses.

As we progress through this book, Nexus will be added to support new agent features. However, simultaneously, the intent will be to keep things relatively simple to teach many of these essential core concepts. In the next section, we'll look at how to quickly use Nexus before going under the hood to explore features in detail.

7.1.1 Running Nexus

Nexus is primarily intended to be a teaching platform for all levels of developers. As such, it will support various deployment and usage options. In the next exercise, we'll introduce how to get up and running with Nexus quickly.

Open a terminal to a new Python virtual environment (version 3.10). If you need assistance creating one, refer to appendix B. Then, execute the commands shown in listing 7.1 within this new environment. You can either set the environment variable at the command line or create a new `.env` file and add the setting.